Data Structures and Algorithms · 21CSC201J

Unit 1 · Session 3 · SLO 1 — Solved Worksheet

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Unit â€“ 1 : Session â€“ 3 : SLO â€“ 1 | Nested Structures

I. Fill in the Blanks

1. inside â€” a nested structure means a structure inside another structure.
2. struct â€” the struct keyword defines the nested structure.
3. member (inner member name) â€” outer.inner.member.
4. multiple (two or more) â€” one dot per nesting level.
5. nested structure.
6. variable â€” memory is allocated when the outer structure variable is declared.
7. inner (nested) â€” B is the inner/nested structure.
8. related (logically grouped) â€” nested structures group related sets of information.
9. FALSE â€” age is not a member of address; the correct path depends on the design, e.g. student.age or student.dob.date.
10. the dot operator applied repeatedly, outer to inner (outer.inner.member).

II. Code Snippets

1. Student/address: struct Address { char city[30]; int pincode; }; struct Student { char name[50]; struct Address addr; }; â€” access: s.addr.city.
2. Employee/salary: struct Salary { float basic, bonus; }; struct Emp { char name[50]; struct Salary sal; }; â€” e.sal.basic.
3. Book/author: struct Author { char name[50]; char country[20]; }; struct Book { char title[50]; struct Author auth; }; â€” b.auth.name.
4. Car/engine: struct Engine { int capacity; char type[20]; }; struct Car { char brand[20], model[20]; struct Engine eng; }; â€” c.eng.capacity.
5. Hospital/contact: struct Contact { char phone[15]; char email[40]; }; struct Patient { char name[50]; int age; struct Contact con; }; â€” p.con.phone.

III. Practice the Code

- struct Date { int day, month, year; };
- struct Employee { char name[50]; int id; struct Date joining; };
- struct Employee e = {"Ravi", 101, {12, 8, 2023}};
- printf("Joined on %d-%d-%d", e.joining.day, e.joining.month, e.joining.year);